

DATASET INSPECTION REPORT

Customer Churn Dataset Inspection

Submitted By

Hudu Yusuf Ibrahim

Course

Business Analytics

Date

August 2026

1. Introduction

Before performing any business analysis, the dataset was inspected to assess its quality, structure, and suitability for analysis. This inspection aimed to identify missing values, duplicate records, incorrect data types, and any other issues that could affect the accuracy of the results.

2. Dataset Overview

The Telco Customer Churn dataset contains customer demographic information, subscription details, billing information, and customer churn status.

The dataset includes variables such as:

- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service
- Internet Service
- Contract Type
- Payment Method
- Monthly Charges
- Total Charges
- Churn Status

The dataset consists of:

- **7,043 rows**
- **21 columns**

The screenshot shows a JupyterLab window titled "week1_customer_churn_analysis" with a "Last Checkpoint: yesterday" status. The interface includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar with icons for file operations and code execution. The code cell contains the following Python code:

```
[2]: import pandas as pd_
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("ggplot")

[3]: df = pd.read_csv("telco-customer-churn.csv")

[4]: df.head()
```

The output of the code is a preview of the first five rows of the "telco-customer-churn.csv" dataset. The table has 15 columns: customerID, gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, MultipleLines, InternetService, OnlineSecurity, ..., DeviceProtection, TechSupport, and SI. The first five rows are as follows:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	SI
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	

Below the table, it indicates "5 rows x 21 columns" and provides a horizontal scrollbar for the data preview.

3. Dataset Structure

The dataset structure was examined using:

`df.shape`

`df.columns`

`df.info()`

The inspection confirmed:

- 7,043 customer records
- 21 variables
- A combination of categorical and numerical variables

```
Jupyter week1_customer_churn_analysis Last Checkpoint: yesterday
File Edit View Run Kernel Settings Help Trusted
JupyterLab Python 3 (ipykernel)

dtype: object

[9]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   customerID            7043 non-null   object
1   gender                7043 non-null   object
2   SeniorCitizen         7043 non-null   int64
3   Partner              7043 non-null   object
4   Dependents            7043 non-null   object
5   tenure                7043 non-null   int64
6   PhoneService          7043 non-null   object
7   MultipleLines         7043 non-null   object
8   InternetService       7043 non-null   object
9   OnlineSecurity        7043 non-null   object
10  OnlineBackup          7043 non-null   object
11  DeviceProtection      7043 non-null   object
12  TechSupport           7043 non-null   object
13  StreamingTV           7043 non-null   object
14  StreamingMovies       7043 non-null   object
15  Contract              7043 non-null   object
16  PaperlessBilling      7043 non-null   object
17  PaymentMethod         7043 non-null   object
18  MonthlyCharges        7043 non-null   float64
19  TotalCharges          7043 non-null   object
20  Churn                 7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

4. Data Types

The dataset contains both categorical and numerical variables.

Examples include:

Variable	Data Type
gender	Object
Contract	Object
PaymentMethod	Object
tenure	Integer
MonthlyCharges	Float
TotalCharges	Float (after conversion)

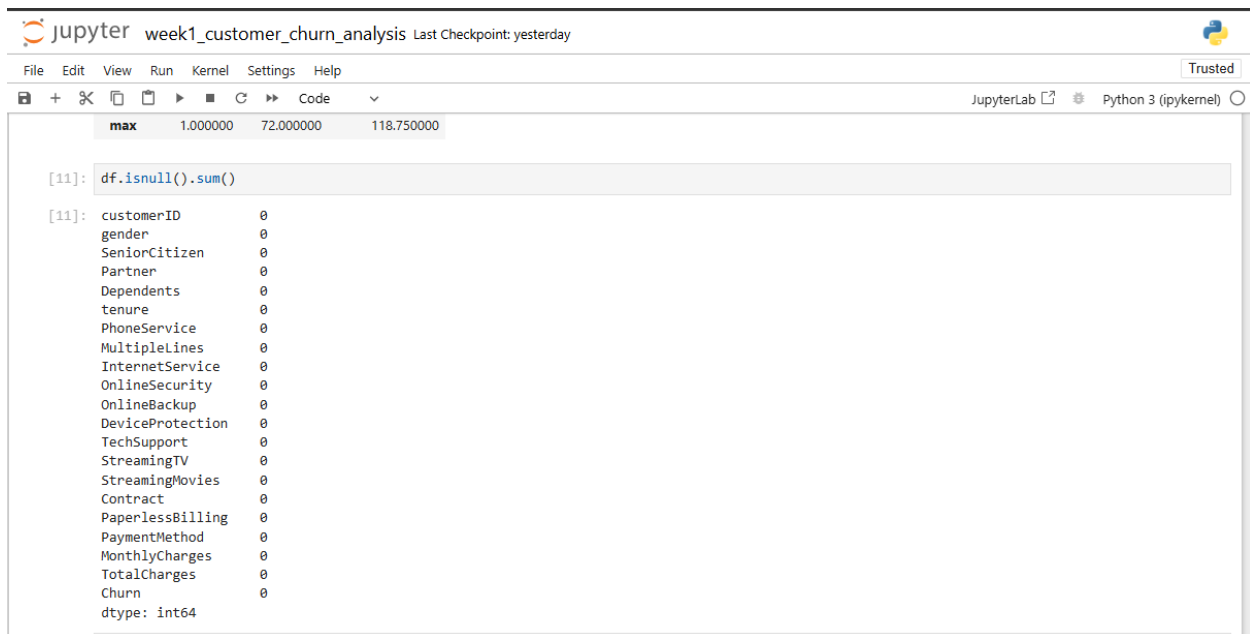
During inspection, **TotalCharges** was initially stored as a text (object) field and was converted to a numeric data type to support numerical analysis.

5. Missing Values

The dataset was checked for missing values using:

```
df.isnull().sum()
```

No significant missing values were found after converting the **TotalCharges** column. Any missing values resulting from the conversion were replaced using the median value of the column.



The screenshot shows a JupyterLab window titled "week1_customer_churn_analysis" with a "Trusted" badge. The interface includes a top bar with "File", "Edit", "View", "Run", "Kernel", "Settings", and "Help" menus. Below the menu bar is a toolbar with icons for file operations and a "Code" dropdown. The main area displays a code cell with the following content:

```
[11]: df.isnull().sum()
```

The output of the cell is a Series of zeros, indicating no missing values:

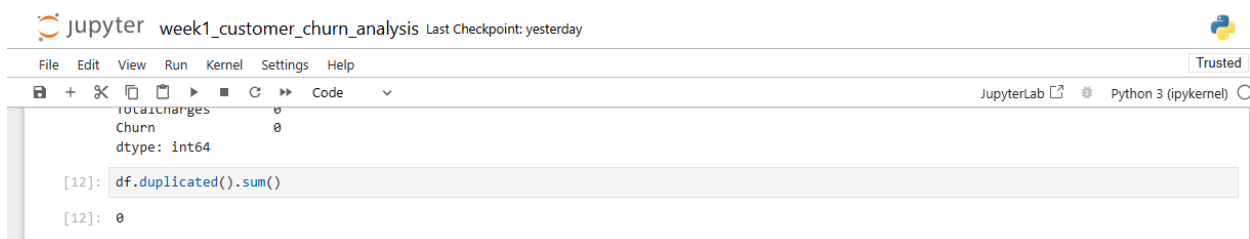
```
[11]: customerID      0
      gender          0
      SeniorCitizen  0
      Partner        0
      Dependents     0
      tenure         0
      PhoneService   0
      MultipleLines   0
      InternetService 0
      OnlineSecurity  0
      OnlineBackup    0
      DeviceProtection 0
      TechSupport     0
      StreamingTV     0
      StreamingMovies 0
      Contract        0
      PaperlessBilling 0
      PaymentMethod   0
      MonthlyCharges  0
      TotalCharges    0
      Churn           0
      dtype: int64
```

6. Duplicate Records

Duplicate records were checked using:

```
df.duplicated().sum()
```

The inspection showed that **no duplicate records** were present in the dataset. This indicates that each customer record is unique and suitable for analysis.



The screenshot shows a JupyterLab window titled "week1_customer_churn_analysis" with a "Trusted" badge. The interface includes a top bar with "File", "Edit", "View", "Run", "Kernel", "Settings", and "Help" menus. Below the menu bar is a toolbar with icons for file operations and a "Code" dropdown. The main area displays a code cell with the following content:

```
[12]: df.duplicated().sum()
```

The output of the cell is 0, indicating no duplicate records:

```
[12]: 0
```

7. Data Cleaning

The following preprocessing steps were performed:

- Converted **TotalCharges** to numeric format.
- Replaced missing values in **TotalCharges** with the median.
- Verified data types after conversion.
- Confirmed there were no duplicate records.
- Confirmed the dataset was ready for analysis.

8. Summary of Inspection

The dataset inspection confirmed that the data was suitable for business analysis.

Key findings include:

- Dataset successfully imported.
- Dataset contains 7,043 customer records.
- Dataset contains 21 variables.
- Data types verified.
- Missing values handled appropriately.
- No duplicate records identified.
- Dataset cleaned and prepared for exploratory data analysis.

9. Conclusion

The dataset inspection process confirmed that the Telco Customer Churn dataset is complete, consistent, and suitable for business analytics. Necessary preprocessing steps were performed to ensure data quality, including converting the **TotalCharges** column to a numeric data type and handling missing values. With these checks completed, the dataset was ready for exploratory analysis and business insight generation.